

Program: Diplomain Communication and Computer Networking	
Course Code: 3325	Course Title: Computer Hardware Lab
Semester: 3	Credits: 1.5
Course Category: Programme core course	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Identify the hardware components of a computer.
- Identify the peripheral devices outside computer.
- Identify the software running on a computer.

Course Outcomes:

On completion of the course, the student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize components of a computer system	8	Applying
CO2	Assemble PC from components	9	Applying
CO3	Interface with parallel ports.	10	Applying
CO4	Interface with serial ports.	10	Applying
	Series Test	4	
	Open Ended Experiments / micro projects	4	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	3			3		
CO3	3						
CO4	3						

3 - Strongly mapped, 2 - Moderately mapped, 1 - Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To familiaize components of a computer system		
M1.01	Familiarization of the components / Cards	2	Applying
M1.02	Familiarization of standard connectors, cords and different ports	3	Applying
M1.03	Familiarization of various computer peripherals	3	Applying
CO2	To assemble PC from components		
M2.01	Assemble PC from different components	7	Applying
	SeriesTest–I	2	
CO3	To interface with parallel ports.		
M3.01	Interfacing LEDs, 7 segment display devices, relays, sensors etc. –	3	Applying
M3.02	Testing of simple logic gates using parallel port	2	Applying
M3.03	Data transfer to the printer by direct access of Data transfer to the printer by direct access of parallel port registers	3	Applying
M3.04	PC to PC data transfer using parallel port	2	Applying
CO4	To interface with serial ports.		
M4.01	Finding the base addresses of COM ports in a system	3	Applying
M4.02	Data acquisition through COM port using ADC chip	3	Applying
M4.03	Serial communication between two computers using BIOS serial port services	2	Applying
	Series Test -II	2	
	Open Ended Experiments / micro projects	4	

Note: Experiments shall be conducted such that all COs are attained.
 Minimum of 6 experiments (excluding open ended experiments / micro projects) shall be performed.
 Compulsory for CIA
 The CIA shall be arranged in third semester by the faculty in charge. The ESE need to be conducted at the end of the third semester.

Text/Reference:

T/R	Book Title / Author
T1	“Computer Hardware: Installation, Interfacing, Troubleshooting and Maintenance” by James K L
R1	“Microprocessors, Pc Hardware and Interfacing” by Mathivanan
R2	“The HCS12 / 9S12: An Introduction To Software And Hardware Interfacing” by Han-Way Huang

OnlineResources

Sl No	Website Link
1	https://vlabs.iitb.ac.in
2	https://vlab.amrita.edu
3	https://www.tutorialsworld.com/computers/pc-motherboard.htm